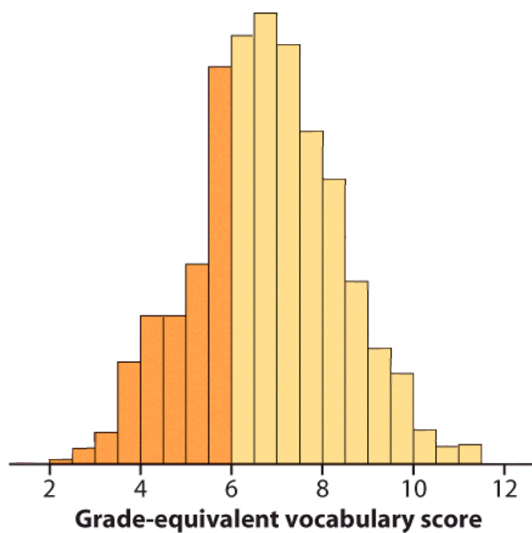


## Lecture 12: The Normal Distribution

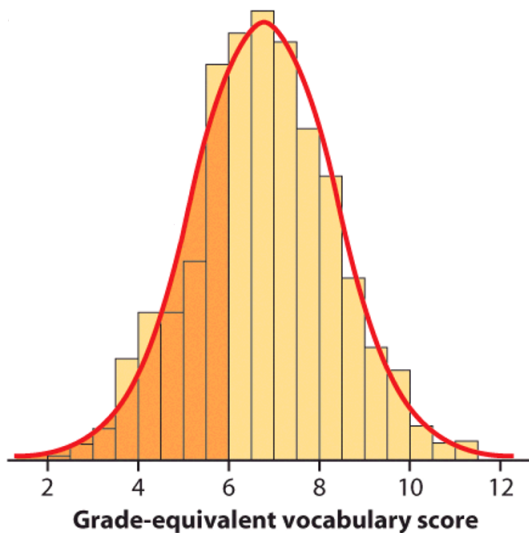
### Continuous Random Variables

Recall: A **continuous** random variable is ...

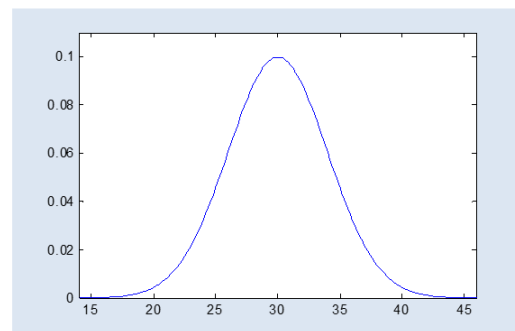
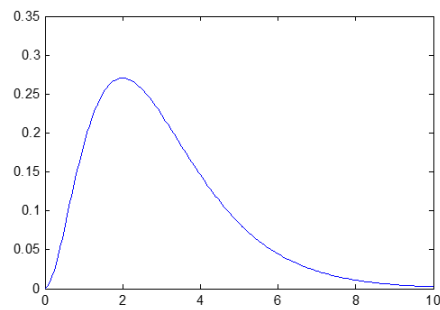
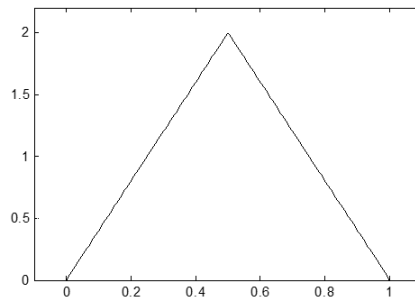
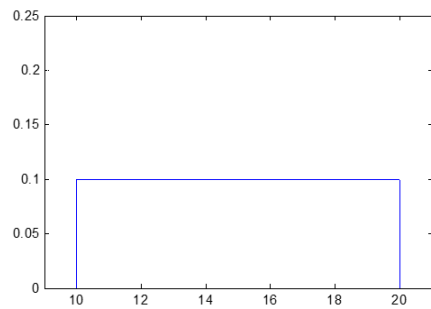
Example: Test scores



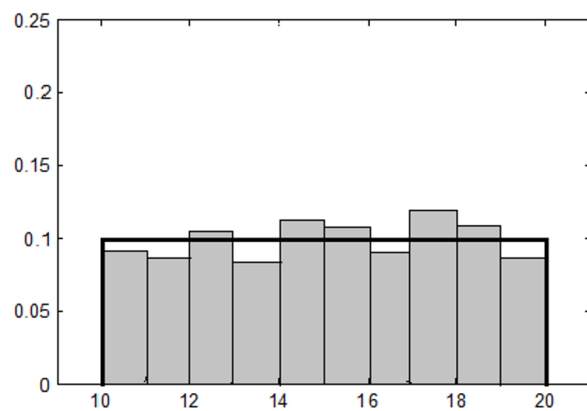
A different approach:



Definition: A **density curve** is ...



Using a density curve to find proportions/probabilities

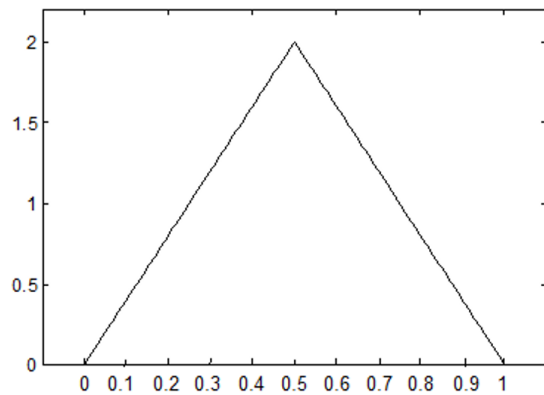


Verify it's a density curve:

Approximate proportion of observations between 14 and 18:

What value corresponds to 20th percentile of dataset?

Another example:

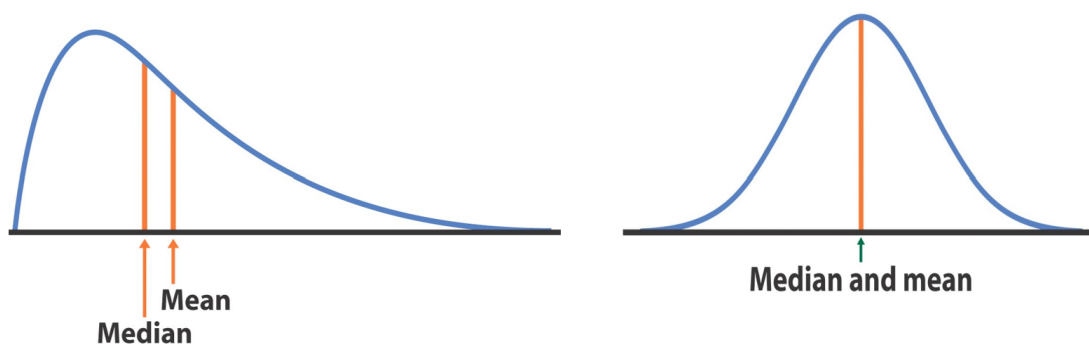


Verify it's a density curve:

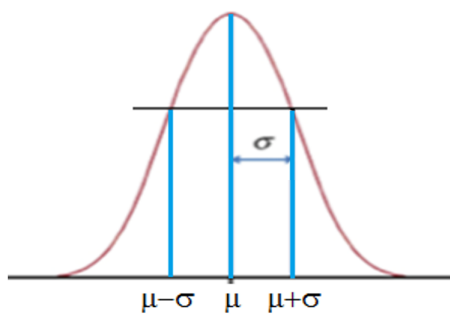
What proportion of observations are between 0.5 and 1?

What proportion of observations are between 0 and 0.2?

Approximating mean and median from a density curve:

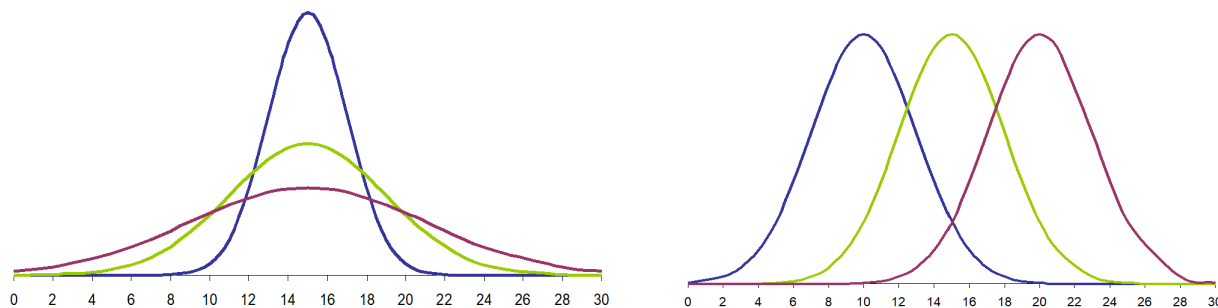


Normal Density Curves

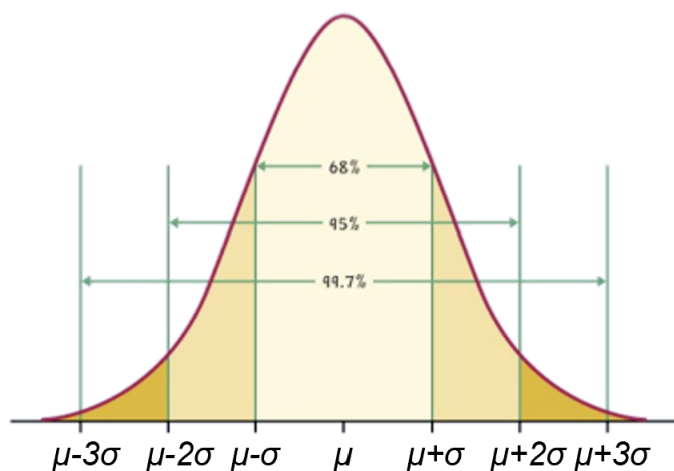


$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad \text{for all } x \in \mathbb{R}.$$

Varying mean and standard deviation:



The 68-95-99.7 Rule



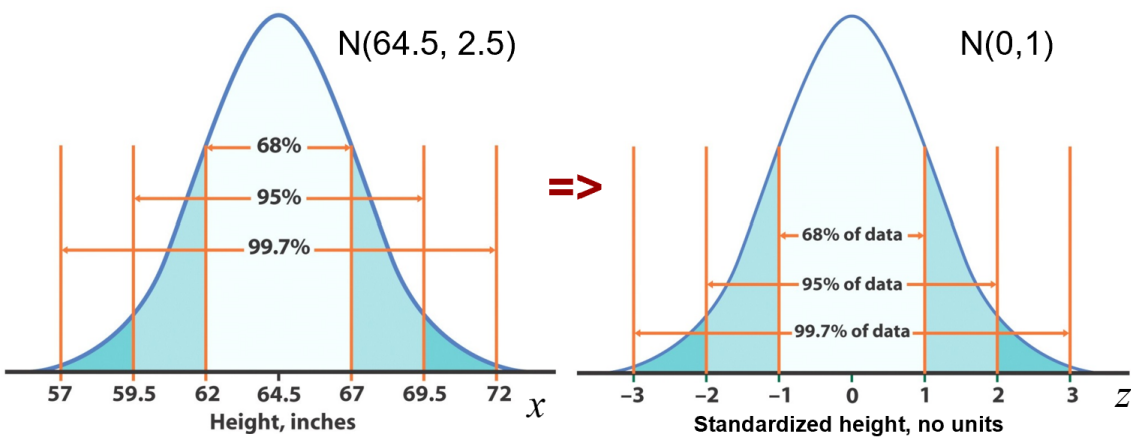
Example: Heights of US women have an approximately normal distribution with mean 64.5 inches and standard deviation 2.5 inches.

- About 68% of US women are between \_\_\_\_\_ and \_\_\_\_\_ inches tall.
- About 95% of US women are between \_\_\_\_\_ and \_\_\_\_\_ inches tall.
- About 99.7% of US women are between \_\_\_\_\_ and \_\_\_\_\_ inches tall.

Example: 2.4 million people took the SAT in 2023. Scores were approximately normally distributed with mean  $\mu = 1025$  and standard deviation  $\sigma = 209$ . Using the 68-95-99.7 rule...

- ... about ..... test takers scored between ..... and .....
- ... about ..... test takers scored between ..... and .....
- ... about ..... test takers scored between ..... and .....

The Standard Normal Density Curve



**z-score**

Example: back to heights of US women

Example: Prices of items at a big-box store are well-approximated by a normal distribution with mean \$20 and standard deviation \$5.

- Calculate the  $z$ -score for an item priced at \$10, then approximate the proportion of items priced below \$10.
- Approximate the proportion of items priced between \$10 and \$25.
- What proportion of items are priced higher than \$22.15?

| $Z$ | Second decimal place of $Z$ |        |        |        |        |        |        |        |        |        |
|-----|-----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|     | 0.00                        | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
| 0.0 | 0.5000                      | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398                      | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793                      | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179                      | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554                      | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915                      | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257                      | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580                      | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881                      | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159                      | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413                      | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643                      | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| ⋮   | ⋮                           | ⋮      | ⋮      | ⋮      | ⋮      | ⋮      | ⋮      | ⋮      | ⋮      | ⋮      |

Table 1.44: A section of the normal probability table. The percentile for a normal random variable with  $Z = 0.43$  has been *highlighted*, and the percentile closest to 0.8000 has also been *highlighted*.

- What price corresponds to the 80<sup>th</sup> percentile of prices of items sold in the store?

Excel functions for calculating normal probabilities and percentiles:

Example: Back to the SAT: 2.4 million people took the SAT in 2023 and Scores were approximately normally distributed with mean  $\mu = 1025$  and standard deviation  $\sigma = 209$ .

- What proportion of test-takers scored below 900?
- What proportion of test-takers scored between 900 and 1200?
- What score marks the top 5% of all scores?

More practice: Body temperatures of healthy humans are distributed nearly normally with mean  $98.2^{\circ}\text{F}$  and standard deviation  $0.73^{\circ}\text{F}$ . What is the cutoff for the highest 10% of human body temperatures?

- (a)  $97.3^{\circ}\text{F}$
- (b)  $99.1^{\circ}\text{F}$
- (c)  $99.4^{\circ}\text{F}$
- (d)  $99.6^{\circ}\text{F}$